

AI Powered Work

Learn AI Prompting Basics

What is a prompt?

A **prompt** is the instruction or question we give to an AI system to tell it what we want.

For example:

"Explain database normalization."

This is a basic prompt.

A better prompt gives more information:

"Explain database normalization to second-year ECE students using a simple example of a student database. Keep the explanation within 300 words."

The second prompt gives the AI **context, goal, audience and expectations**.

Microsoft identifies four important elements of an effective Copilot prompt:

Element	Meaning	Example
Context	Background information	"I am teaching second-year ECE students."
Goal	What you want	"Explain normalization."
Source	Information AI should use	"Use the following notes..."
Expectations	How the answer should look	"Give a table and a simple example."

Better input → Better output

Write Effective Prompts to Achieve Optimal Results

Microsoft recommends prompts that are:

- Clear
- Specific
- Contextual
- Relevant
- Concise

Suppose you want AI to create questions for students.

Weak prompt

Give me questions on C.

Effective prompt

Create 10 multiple-choice questions on C operators for first-year engineering students. Include +, -, *, /, % and operator precedence. Provide four options and mark the correct answer.

The second prompt tells AI:

WHO? → First-year engineering students

WHAT? → MCQs

TOPIC? → C operators

HOW MANY? → 10

FORMAT? → 4 options + answer

Prompting for more than Code

AI prompting is **not only for programming**.

You can use AI to work with:

Text

Summarize this article in 5 bullet points.

Education

Create 10 MCQs on DBMS normalization for second-year students.

Email

Rewrite this email in a professional and polite tone.

Presentation

Create a 10-slide presentation structure on renewable energy.

Data

Analyze these marks and identify students who need remedial classes.

Image/document understanding

Extract the important points from this document.

Brainstorming

Give me five project ideas using machine learning for agriculture.

Translation

Translate this announcement into Tamil.

Get Started with Microsoft Copilot

Microsoft Copilot is **Microsoft's AI assistant** that can help users with tasks such as:

- Asking questions
- Writing
- Summarizing
- Brainstorming
- Understanding information
- Creating content

Instead of searching manually for every small task, you can describe the task in natural language.

Application	AI assistant	What it helps with
Microsoft Word	Copilot	Writing, rewriting, summarizing documents
Microsoft PowerPoint	Copilot	Create presentations from prompts
Microsoft Excel	Copilot	Analyze data, formulas, charts
Microsoft Outlook	Copilot	Draft/reply to emails, summarize threads
Canva	Magic Studio	Presentations, posters, designs, images
Google Docs/Gmail	Gemini	Writing, summarizing, emails
Adobe apps	Firefly	Image generation/editing
Notion	Notion AI	Notes, summaries, organization
Zoom	AI Companion	Meeting summaries and action items
GitHub Copilot	Copilot	Programming

Work Smarter Using Copilot

This topic focuses on using AI to increase productivity.

Traditional approach	With Copilot
Read document ↓ Understand document ↓ Write summary ↓ Create email ↓ Prepare presentation	Document ↓ Copilot ↙ ↓ ↘ Summary Email Presentation

Example

Document

"Summarize this 20-page document in one page."

Instead of manually reading everything, Copilot helps extract the useful information

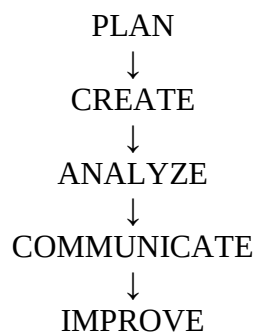
Presentation

"Convert these notes into a 10-slide presentation."

Copilot for All the Ways You Work

This topic expands the idea of **Copilot beyond simple question-answering**.

AI can assist at different stages of work:



Example

Preparing a course Plan

Create a 5-unit course plan for Java.

Create lecture notes for Unit 1.

Analyze

Analyze these student marks and identify weak areas.

Thus, Copilot becomes a **work assistant**, rather than simply a chatbot.

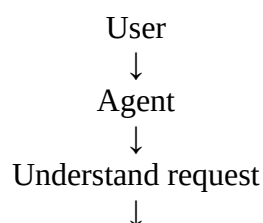
Build Smarter, Faster: Custom AI Agents with Copilot Studio

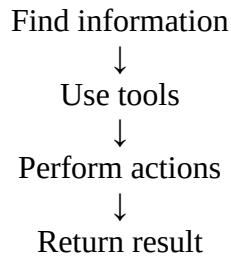
What is an AI agent?

A normal chatbot mainly:

User → Question → AI → Answer

An **AI agent** can be designed to:





Example 1: College Admission Agent

Imagine a college creates an agent called:

"College Admission Assistant"

A student asks:

"What documents are required for admission?"

The agent can:

1. Understand the question.
2. Search the college's approved information.
3. Retrieve the required documents.
4. Give the answer.

Another student asks:

"What is the last date?"

The agent can retrieve the relevant admission information.

Example 2:

"Find suitable restaurants near me for dinner and check availability."

An agent can:

1. Understand your requirement
2. Search restaurants
3. Check information
4. Check reservation availability
5. Present suitable options

Examples of AI Agents

AI Agent	What it does
College Admission Agent	Answers eligibility, courses, fees, documents, admission dates
Teaching Agent	Creates lesson plans, quizzes, assignments, explanations
Student Study Agent	Answers questions, creates study plans, gives practice questions

Hospital Appointment Agent	Finds available doctors, schedules appointments, sends reminders
Shopping Agent	Understands requirements, compares products, recommends suitable products
Travel Agent	Plans itinerary, finds flights/hotels, organizes a trip
HR Recruitment Agent	Screens resumes, checks qualifications, schedules interviews
Email Agent	Reads emails, identifies important messages, drafts replies
Banking Support Agent	Answers account questions, checks permitted information, helps with requests
Customer Service Agent	Handles customer questions, checks orders, creates support tickets
IT Helpdesk Agent	Diagnoses common computer problems and suggests/executes fixes
Data Analysis Agent	Takes a dataset, analyzes it, creates charts/reports and explains results

Custom Prompts in Copilot Studio

In Copilot Studio, you can create custom prompts that control how the AI performs a particular task.

Example

Task: Create an email to a faculty member.

Custom Prompt:

"Act as a college student. Write a polite and professional email to a faculty member. Keep the email short and use simple English."

Input:

"I was absent yesterday because I was sick."

Copilot Output:

"Dear Sir/Madam,
I was unable to attend college yesterday as I was unwell. Kindly excuse my absence.
Thank you."

Transform Your Work with Agents

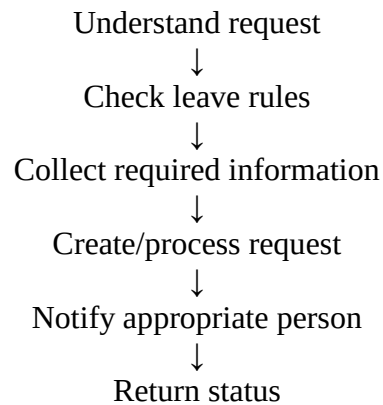
This is the next level: AI agents can become active participants in workflows.

Example: Student Leave Agent

Student:

"I need leave tomorrow because I have a medical appointment."

Agent could potentially:



Summary

Step 1 – Prompt

"Analyze these marks."

Step 2 – Better prompt

"Analyze the marks of 30 CSE students and identify students scoring below 50%."

Step 3 – Copilot

Ask Copilot to:

"Summarize the performance and suggest remedial activities."

Step 4 – Custom prompt

A custom prompt is simply a set of instructions you give an AI once, so that it follows your preferred way of responding.

Step 5 – Agent

Build a **Student Academic Support Agent**.

How to build an agent from existing AI Model

Step 1 — Decide the job of the agent

First decide exactly what you want it to do.

For example:

"Answer students' questions about college admissions."

Step 2 — Choose an existing AI model

You **don't** create the AI model yourself.

You use an existing model, such as a GPT model, as the **brain**.

Existing AI Model



Step 3 — Give the agent instructions

This is your **custom prompt**.

For example:

“You are a College Admission Assistant. Answer students politely and in simple English. Give information only from the college's official admission information. If you don't know the answer, say that the student should contact the admission office.”

This tells the AI **how to behave**.

Step 4 — Give it college knowledge

Now provide the information the agent needs.

For example:

College Admission Information

- Courses
- Eligibility
- Fees
- Documents
- Admission Procedure
- Important Dates
- Contact Details

Custom prompt = instructions

College documents = knowledge

Step 5 — Add tools if necessary

Suppose you want the agent to do more than answer questions.

For example:

“Check whether my application has been submitted.”

The agent may need access to an **admission database/system**.

So you connect an appropriate tool or system.

Step 6 — Test the agent

Ask it questions such as:

“Can I apply for CSE?”

“What documents are required?”

“What is the fee?”

“When does admission start?”

Then check whether the answers are correct.

Step 7 — Make it available to students

Finally, you provide an interface where students can interact with it.

The most important point

You are **not building a new AI model**.

You are building/configuring a **specialized AI application (agent)** around an existing model.

GPT model → brain

Custom prompt → instructions

College documents → knowledge

Connected systems → tools

All together → College Admission Agent

Assignment 2

No.	Assignment	Student Task
1	AI as a Study Assistant	Choose one subject topic and use an AI tool to explain it at three levels: school student, first-year college student, and advanced

		student. Compare the responses.
2	Prompt vs. Effective Prompt	Give an AI tool a simple prompt such as “ <i>Explain DBMS.</i> ” Then create an improved prompt by specifying role, context, goal, and expected output . Compare both answers.
3	AI for Daily College Work	Select one college task such as preparing an email, lesson plan, presentation outline, quiz, or meeting summary. Use AI to complete it and explain how AI saved time.
4	Role-Based Custom Prompt	Create a custom prompt that makes AI act as a specific role: Student / Career Advisor / Programming Tutor / College Admission Assistant . Test it with 5 questions and record the responses.